

Curcumin exerts antidiifferentiation effect through AMPKalpha-PPAR-gamma in 3T3-L1 adipocytes and antiproliferatory effect through AMPKalpha-COX-2 i...



Curcumin has been reported to have the potential to prevent obesity as well as cancers. The downstream targets regulated by AMP-activated protein kinase (AMPK) for inhibiting adipocyte differentiation or cancer cell proliferation of curcumin were investigated. The activation of AMPK by curcumin was crucial for the inhibition of differentiation or growth in both adipocytes and cancer cells. Stimulation of AMPK by curcumin resulted in the down-regulation of PPAR (peroxisome proliferator-activated receptor)-gamma in 3T3-L1 adipocytes and the decrease in COX-2 in MCF-7 cells. Application of a synthetic AMPK activator also supported the evidence that AMPK acts as an upstream signal of PPAR-gamma in 3T3-L1 adipocytes. In cancer cells, AMPK was found to act as a regulator of ERK1/2, p38, and COX-2. Regulation of AMPK and its downstream targets such as PPAR-gamma, Mapkinases, and COX-2 by curcumin appears to be important in controlling adipocytes and cancerous cells.